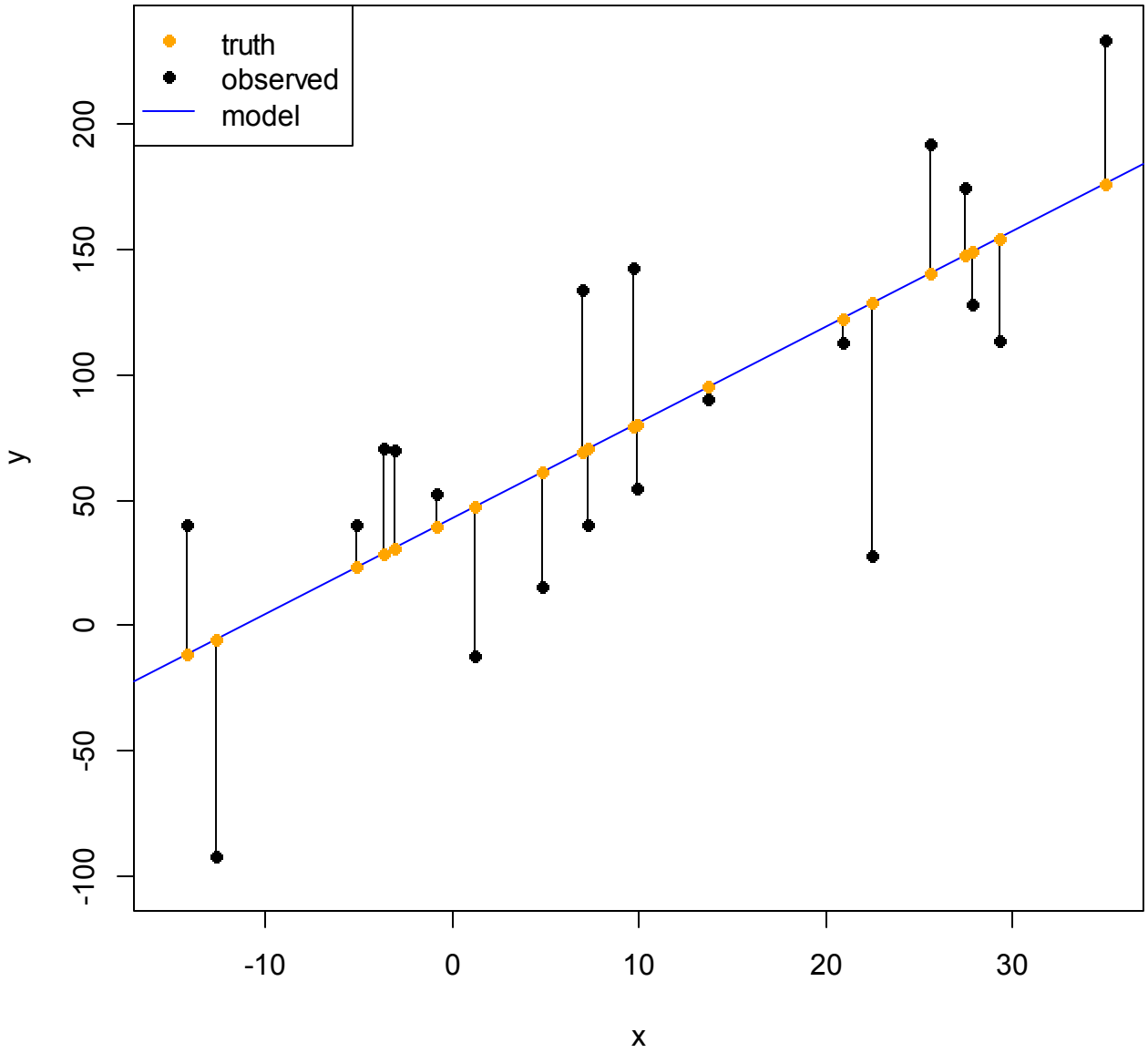
Today

- NBBG model (stochasticity!) finish off
- LOOIC
- Parallel processing (in R)
- Types of uncertainty/stochasticity
  - process uncertainty
  - observation uncertainty
- Estimating sources of uncertainty
  - design, model, training
- State space models

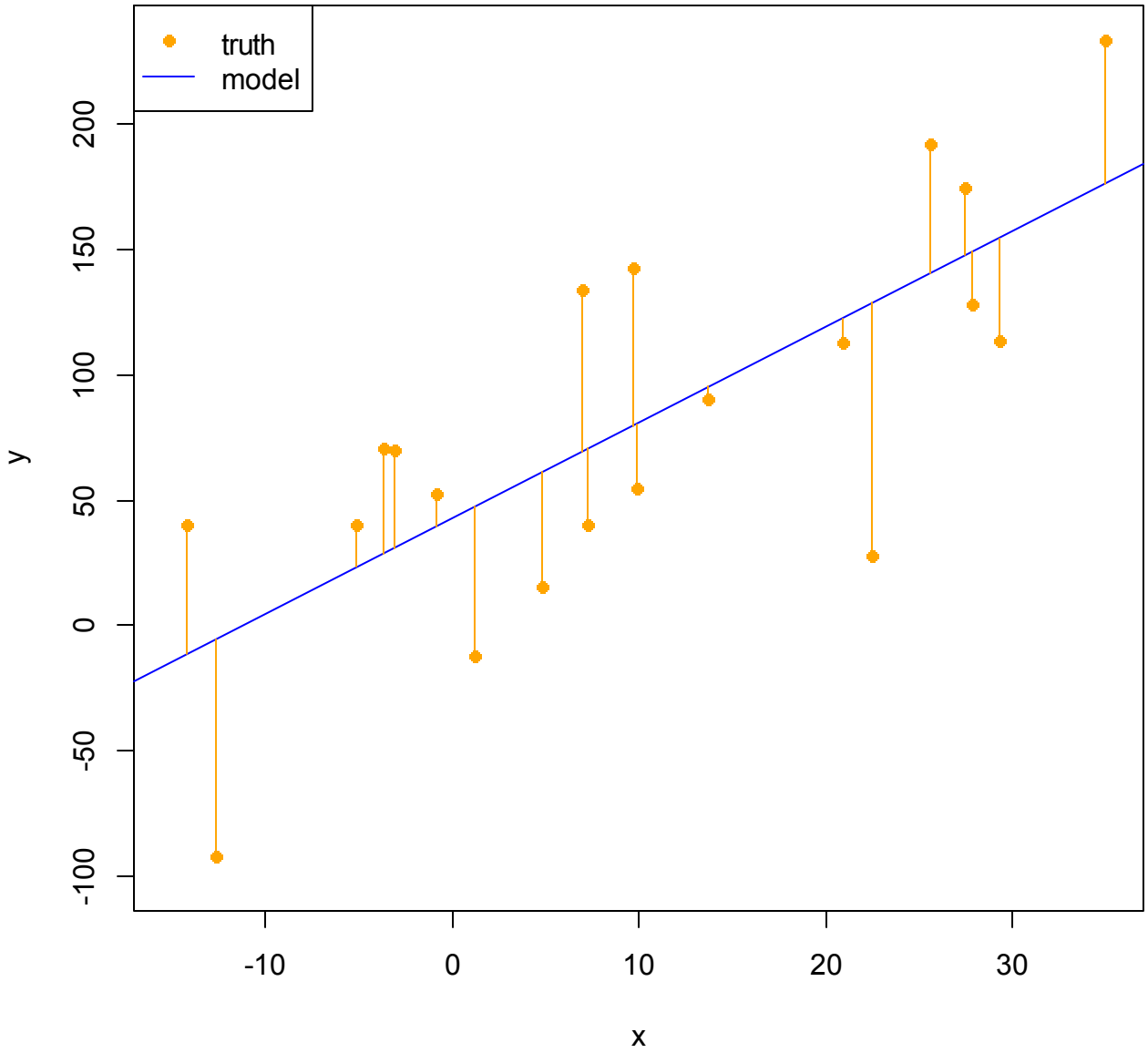
Types of uncertainty



Observation uncertainty

Model is truth, observations are wrong:
measurement error

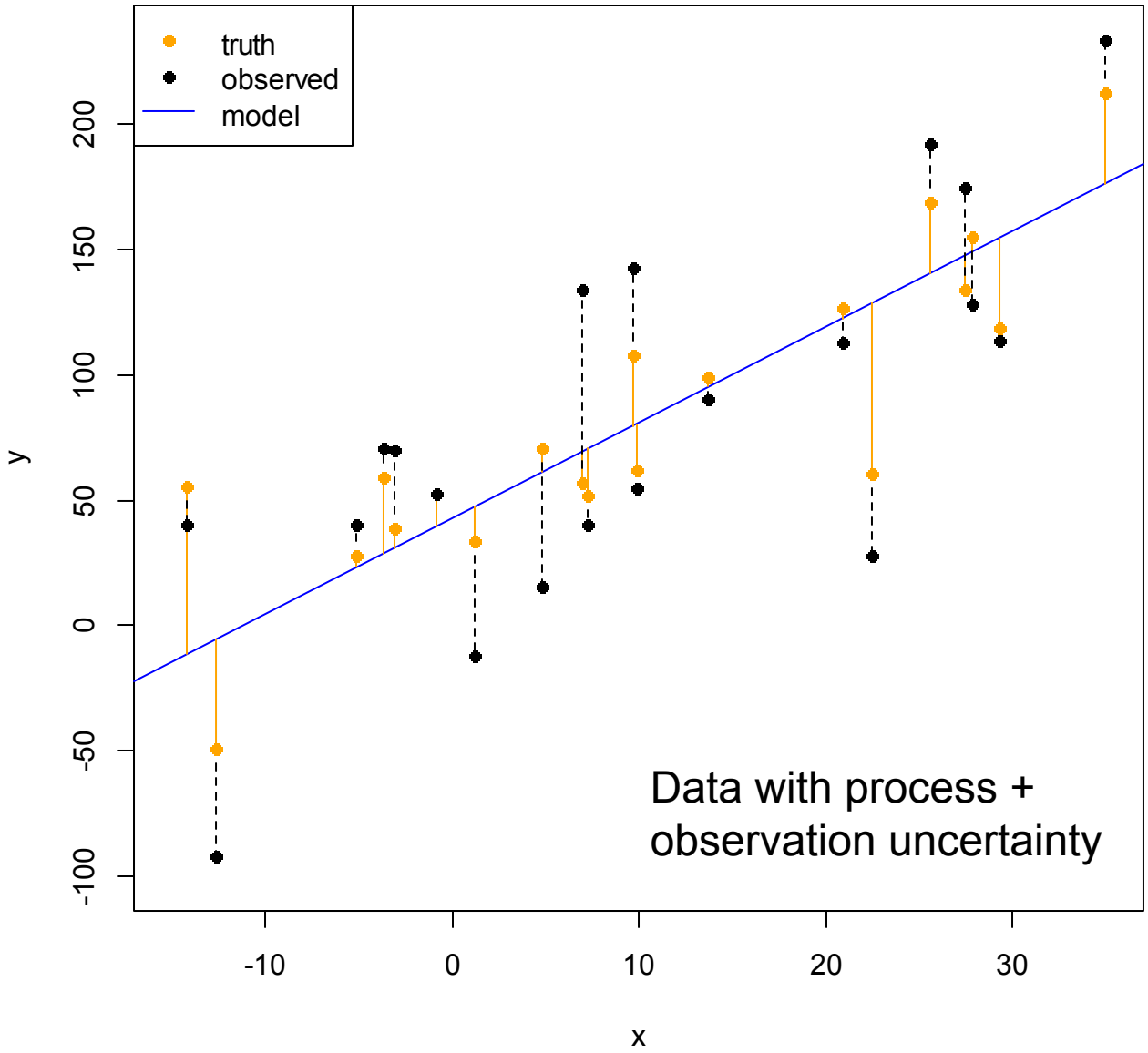
Types of uncertainty



Process uncertainty

Observations are truth, model is wrong

Types of uncertainty



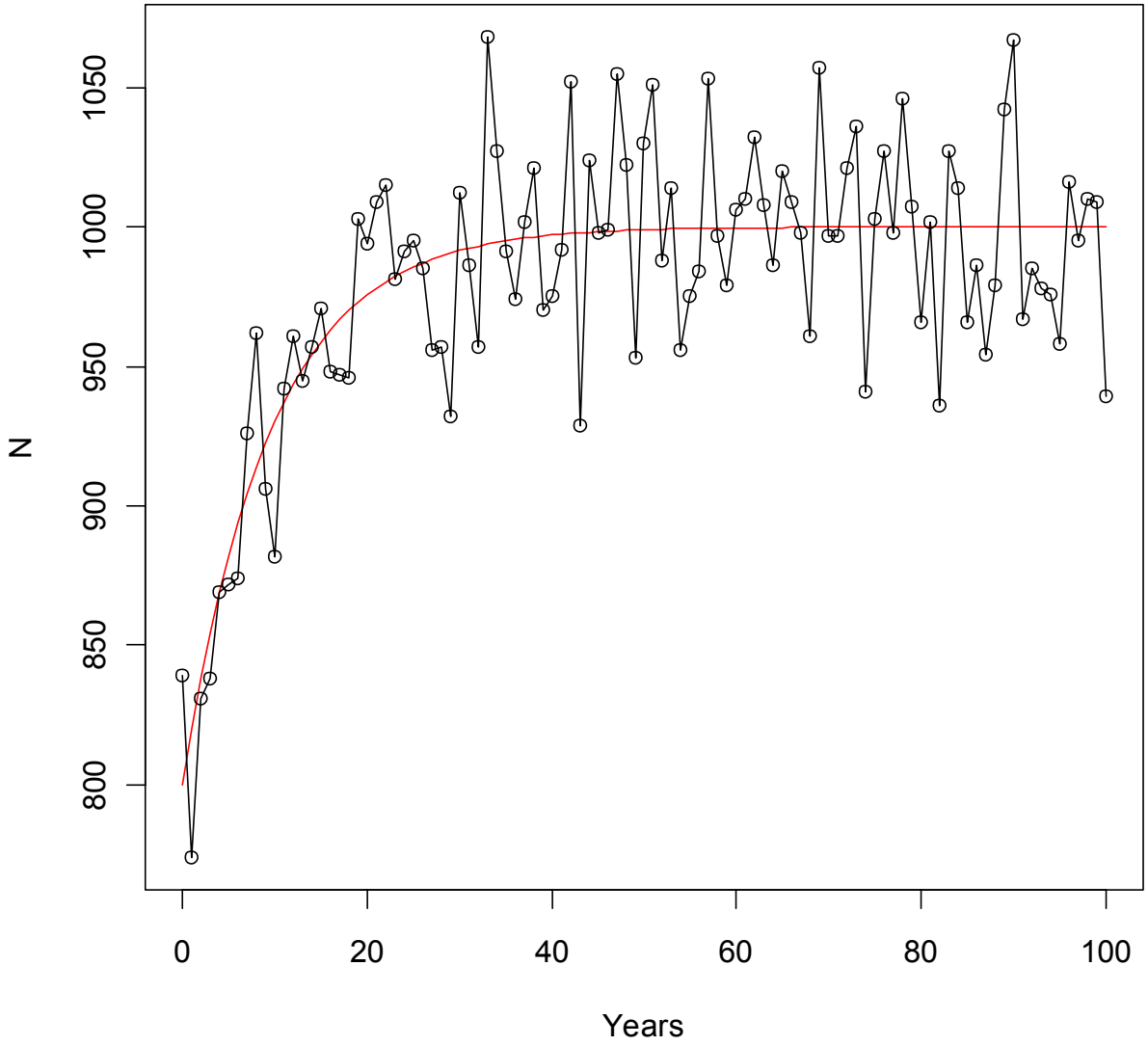
Process uncertainty

Deviations from the model due to unknown processes

Observation uncertainty

Deviations from the truth due to measurement error

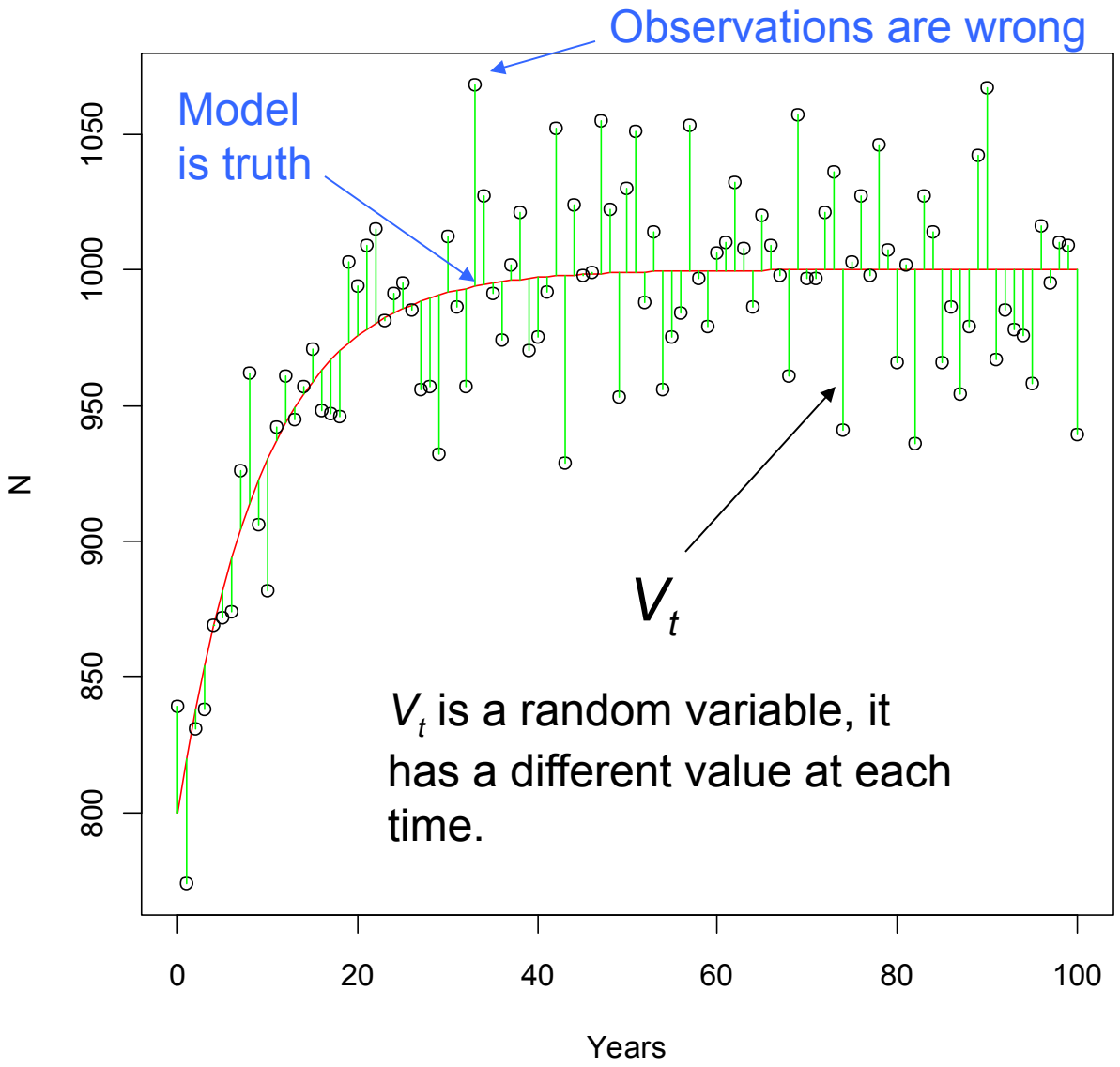
Types of uncertainty



Observation uncertainty

Model is truth, observations are wrong:
Measurement error

Types of uncertainty



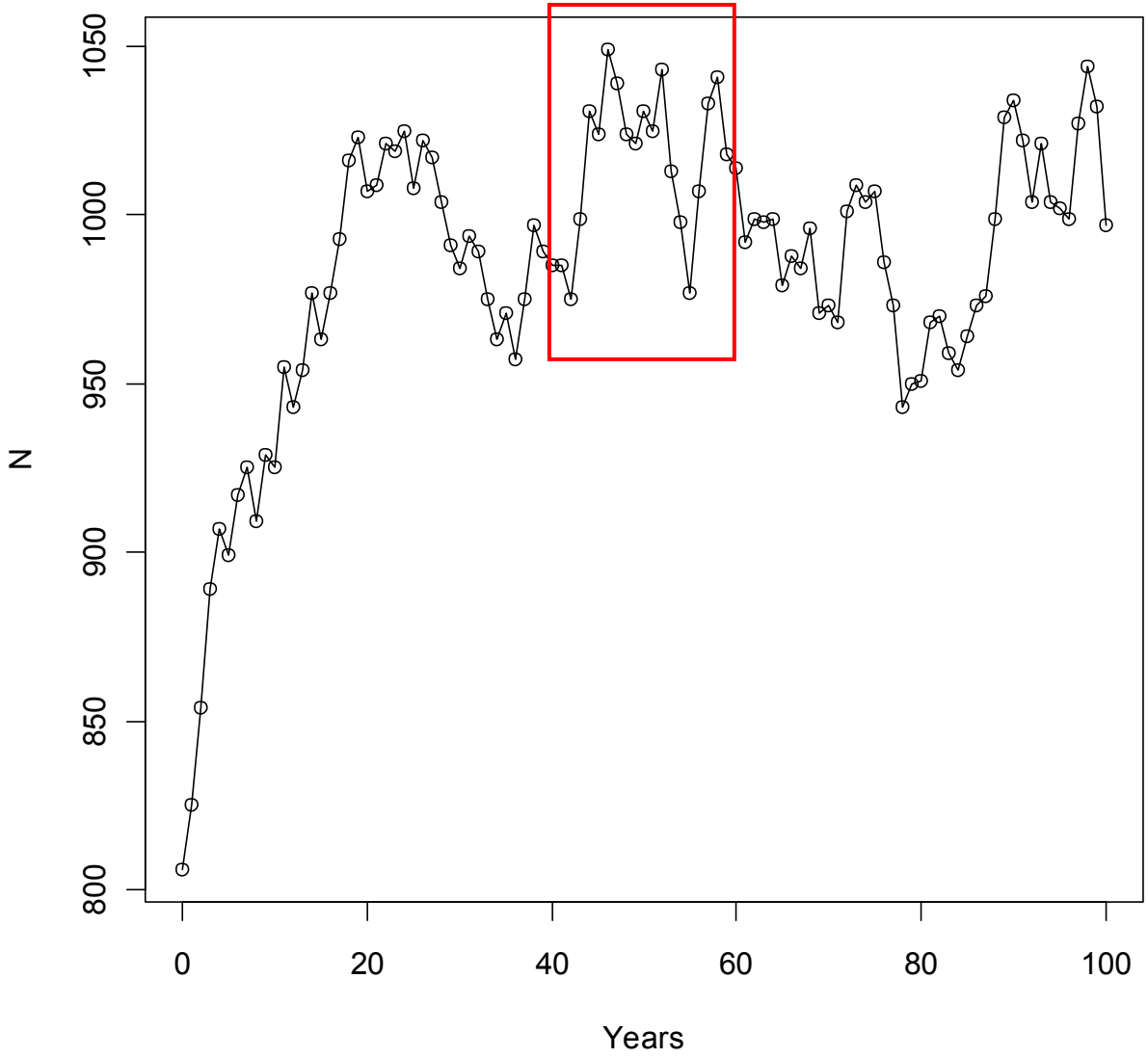
Observation uncertainty

Model is truth, observations are wrong:
Measurement error

$$N_{t+1} = i + sN_t$$

$$N_{obs,t} = N_t + V_t$$

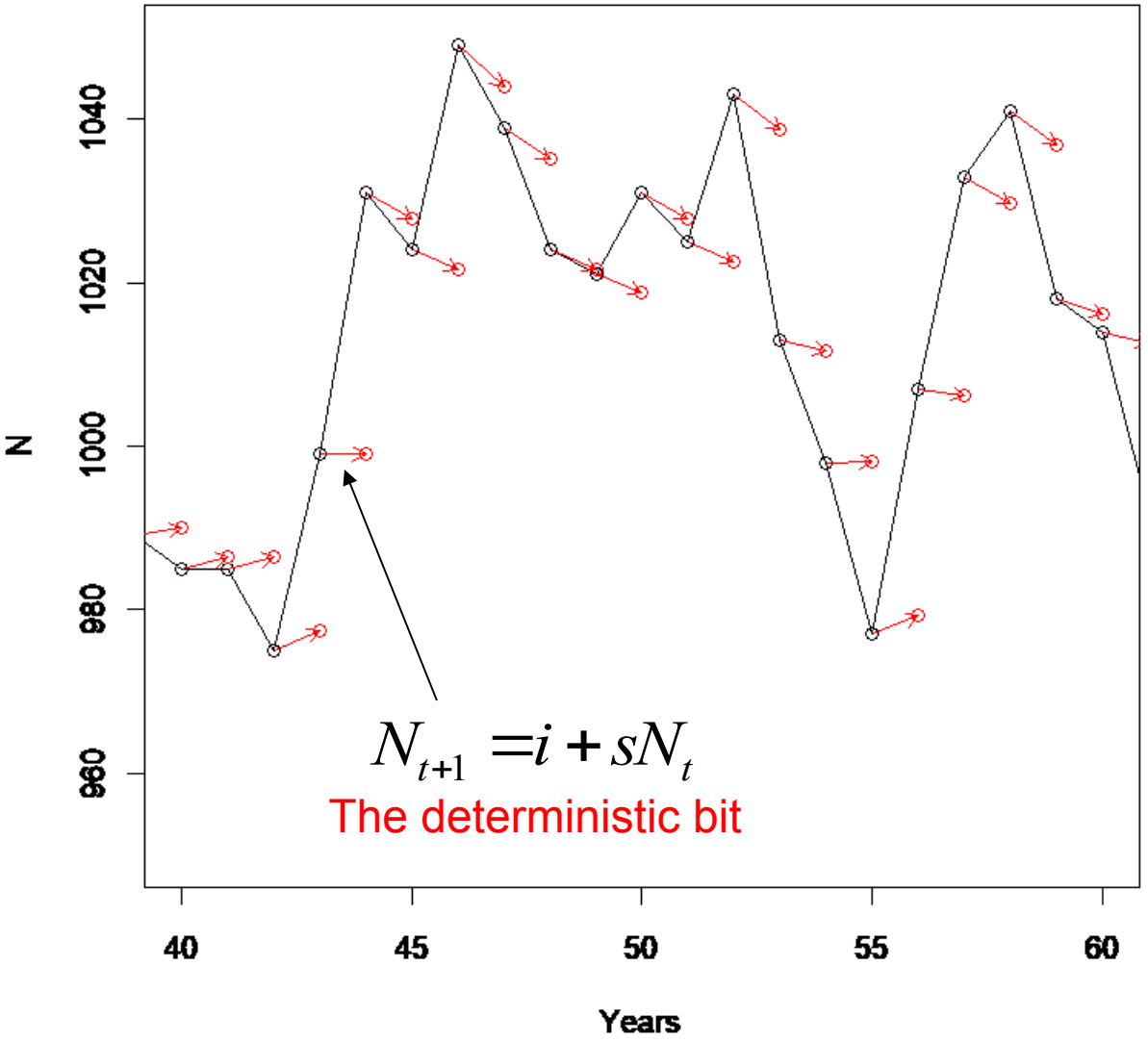
Types of uncertainty



Process uncertainty

Observations are truth, model is wrong

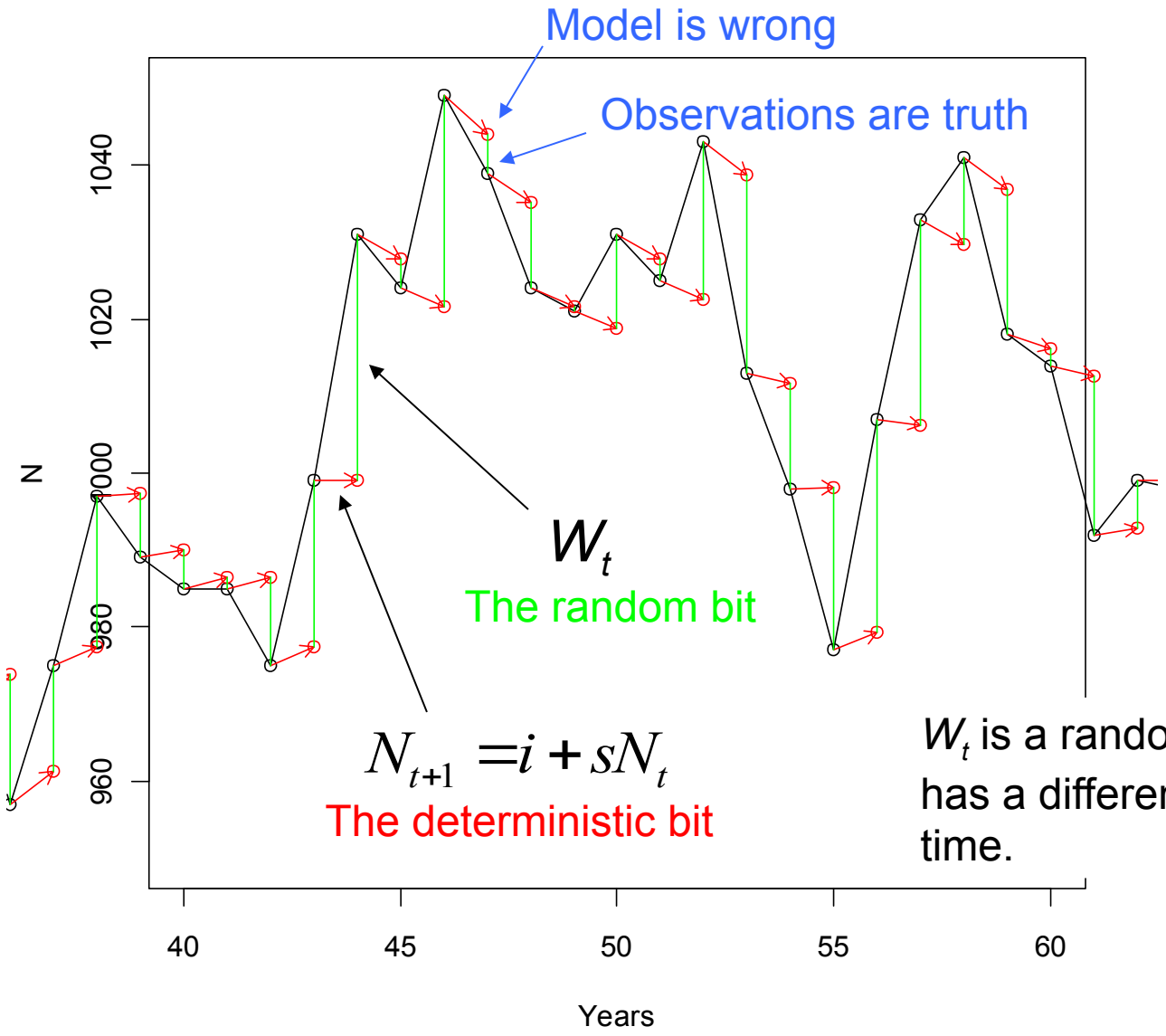
Types of uncertainty



Process uncertainty

Observations are truth, model is wrong

Types of uncertainty



Process uncertainty

Observations are truth, model is wrong

$$N_{t+1} = i + sN_t + W_t$$

W_t is a random variable, it has a different value at each time.